program is throwaway (some are!), then write the API today that you will be happy to use in six months' time.

So, how do you implement an API like `Cézanne` on an immediate-mode API? The following listing shows an implementation of a retained-mode `Cézanne` API that can draw and animate circles.

Listing 11.7. A retained-mode circle-drawing API for `canvas` called `Cézanne`

```
Cézanne = do ->

  seconds = (n) -> n*1000
  framesPerSecond = 30
  tickInterval = seconds(1)/framesPerSecond

  circlePrototype =
    radius: (radius) ->
      @radius = radius
      this
    color: (hex) ->
      @hex = hex
      this
    position: (x, y) ->
      @x = x
      @y = y
      @context.beginPath()
      @context.fillStyle = @color
      @context.arc @x, @y, @radius, (Math.PI/180)*360, 0, true
      @context.closePath()
      @context.fill()
      this
    animatePosition: (x, y, duration) ->
      @frames ?= []
      frameCount = Math.ceil seconds(duration)/tickInterval
      for n in [1..frameCount]
        if n is frameCount
          do =>
            frame = n
            @frames.unshift =>
              @position x, y
        else
          do =>
            frame = n
            @frames.unshift =>
              @position x/frameCount*frame, y/frameCount*frame

  sceneProtocol =
```